

The Occurrence Birth-Death Diffusion Process: Unraveling Diversification Histories with Fossils and Heterogeneous Rates

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ABSTRACT

Phylogenetic diversification analyses often overlook the wealth of information contained in fossils that cannot be directly integrated into phylogenetic trees (called **uncoded** occurrences), as well as fine-grained variations in speciation and extinction rates across lineages. Addressing both of these shortcomings, we develop the Occurrence Birth-Death Diffusion (OBDD) process. This process incorporates the sampling of **uncoded** occurrences by extending the recently developed Fossilized Birth-Death Diffusion (FBDD) process, thereby taking advantage of **a clade's entire observed fossil record**, to infer its evolutionary history. To mitigate the computational challenges associated with the OBDD process, we employ Bayesian data augmentation methods. We illustrate our approach with an empirical analysis of cetacean diversification. Our analyses show that the integration of fossils is essential to recover diversification rates and richness trajectories. More specifically, all types of fossils are highly informative about past diversification dynamics, but **uncoded** occurrences do not provide significant additional information when the phylogeny already contains many fossils. We make the OBDD available in the Julia package Tapestry.

Key words: Occurrence Birth-Death Diffusion, Diversification, Phylogenies, Fossils, Rate Heterogeneity, Cetaceans

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INTRODUCTION

20 The study of diversification histories sheds light on the patterns and processes that drive
21 the emergence, extinction, and evolution of species. Initiated by extensive
22 paleontological research on the fossil record, the field has seen significant advancement
23 with the introduction and refinement of phylogenetic diversification models. These
24 models represent diversification as a birth-death (BD) process, where each lineage can
25 give birth or go extinct at given rates (commonly denoted as λ for speciation and μ for
26 extinction) (Nee et al., 1994; Stadler, 2009, 2010). Early iterations of these models were
27 developed for phylogenies of extant-only taxa and assumed constant rates over time and
28 across lineages, but they have progressively incorporated more complexity, e.g. allowing
29 for temporal and among-lineage variation in rates and dependencies on traits or
30 environmental factors (Stadler, 2013; Morlon, 2014; Morlon et al., 2024).

31 Recently, substantial progress has been made in integrating direct fossil evidence
32 into phylogenetic diversification analyses by adding sampling processes to the
33 birth-death process, thus leveraging the extensive information contained in
34 paleontological databases. This resulted in the development of the Fossilized
35 Birth-Death (FBD) model, which integrates into a phylogeny fossils with adequate
36 morphological or taxonomic information (Stadler, 2010; Heath et al., 2014). However, this
37 information is not available for most fossils, leading traditional FBD models to ignore a
38 large portion of the fossil record. In fact, due to the effort required to compile these data,
39 researchers often include only one fossil per morphospecies in matrices of morphological
40 characters (Wright et al., 2022). In response, the Occurrence Birth-Death (OBD) model
41 was developed to include [uncoded](#) occurrences, that is, the remaining fossils that lack
42 the publicly compiled morphological or taxonomic information that would allow them
43 to be placed in a phylogeny (Vaughan et al., 2019; Gupta et al., 2020; Andréoletti et al.,

2022). From now on, we will use “coded fossil” to designate those placed in the tree, and “uncoded occurrence” for those that are not. Although uncoded occurrences are not as informative as coded fossils, variation in their number over time can reflect variation in a clade’s total species richness and inform the underlying speciation and extinction rates. This is analogous to the complementary information provided by genome sequences (analogous to fossil characters) and incidence data (analogous to fossil occurrences in paleobiology; note that fossil occurrences also carry spatial and assemblage information that our model does not currently use) to understand pathogen dynamics, and, in fact, the same phylodynamic tools are used in this context (MacPherson et al., 2022; Stadler et al., 2024). Analyses of the OBD model in the context of pathogen phylodynamics have shown that sampling times (incidence data) often play a key role in driving inference when sequence diversity is low; in contrast, in scenarios of rapid evolution, sequence data provide the dominant source of information (Featherstone et al., 2023). This suggests that occurrences will be particularly useful when the placement of fossils in the phylogeny is uncertain, for example due to few or similar morphological characters. The OBD model was first implemented by Vaughan et al. (2019) in the phylogenetic software BEAST2 (Bouckaert et al., 2019) with a particle filter algorithm. Next, Gupta et al. (2020) developed a faster algorithm and Andréoletti et al. (2022) extended the model to account for the piecewise constant rate variation and implemented it in RevBayes (Höhna et al., 2016). Both implementations allow full joint inference of phylogenetic trees and diversification (or epidemiological) parameters. However, their application to large trees is hampered by computational limitations, primarily due to a numerically intensive integration step required for the likelihood computation. A fast approximation of this model using a method called TimTam (Zarebski et al., 2025) has been implemented in Beast2 but is currently limited to a scenario with lineage removal at sampling, which makes sense in the case of sampling pathogen genomes, but is not realistic in the case of sampling fossils. In addition, all current implementations of the OBD model assume

homogeneous diversification rates across lineages.

That different clades undergo heterogeneous diversification dynamics have been studied in paleontology for decades (e.g., Sepkoski, 1981). Substantial progress has also been made in developing phylogenetic diversification models that accommodate rate heterogeneity, in which rates can differ between portions of the tree, either due to rare clade-wide shifts (Rabosky, 2014; Mitchell et al., 2019; Maddison et al., 2007; Caetano et al., 2018), or more gradual branch-specific changes (Maliot et al., 2019; Quintero et al., 2024). In particular, Quintero et al. (2024) recently developed the Birth-Death Diffusion (BDD) process, in which diversification rates diffuse along branches according to a geometric Brownian motion process. The BDD model does not integrate a fossil sampling process, which prevents reliably inferring heterogeneous extinction rates. However, it can incorporate information from the fossil record by constraining extinction trajectories independently inferred for the clade (or subclades) from fossil data. The model has now been extended into the Fossilized Birth-Diffusion (FBDD) process (Quintero et al., 2025), which accommodates fossils placed in the phylogeny. Both the BDD and FBDD models have been implemented in the Julia package Tapestry (Quintero and Andréoletti, 2020), with an inference technique called Bayesian data augmentation (see Methods and Tanner and Wong (1987); Maliot and Morlon (2022)). Diversification analyses in Tapestry operate on fixed phylogenies, but Bayesian data augmentation has also been used for full inference of phylogeny and diversification processes (Barido-Sottani and Morlon, 2023; Chabrol et al., 2025). Data augmentation circumvents some mathematical and computational challenges of computing likelihoods of incomplete observed data; it also produces posterior samples of the full diversification histories, i.e. complete phylogenies including unsampled lineages and, in the case of diffusion models, heterogeneous diversification rates evolving along all (observed and unobserved) branches. From this posterior sample of complete phylogenies, one can directly obtain the distribution of estimated diversity-through-time trajectories for a

clade, as well as study in detail the patterns of speciation and extinction at the lineage level. For example, using the BDD model, Quintero et al. (2024) observed an imbalanced speciation process in mammals wherein most lineages undergo a progressive decline in speciation rates before going extinct - notably at the K-Pg mass extinction event - while a few lineages preserve a high speciation potential and drive the diversification of their clade. The selective extinction of lineages with low speciation rates could only be captured by taking into account the rates of the unobserved lineages. Similarly, using the FBDD model, Quintero et al. (2025) studied diversity trajectories across 27 radiations and used data-augmented histories to show that diversity expansions and slowdowns are modulated by speciation, while declines are primarily governed by extinction rates, partly consistent with previous findings at the level of higher taxa by Gilinsky and Bambach (1987). The current implementation of FBDD aims to correct for a mismatch between the FBD process, which assumes a Poisson sampling of fossils, and paleontological databases, in which each morphospecies is typically represented by a single entry, even though many are known from multiple specimens (Wright et al., 2022). To this end, taxonomical information can be used in the FBDD inference to add uncoded occurrences corresponding to the morphospecies included in the phylogeny. Although this reduces the mismatch mentioned above, the model does not incorporate a specific sampling process for occurrences and ignores occurrences that do not belong to included morphospecies.

Building on these recent developments, we introduce the Occurrence Birth-Death Diffusion (OBDD) model, and implement it using Bayesian data augmentation in Tapestry, together with its simpler homogeneous variant, the Occurrence Birth-Death (OBD) model. These models combine the strengths of their predecessors while tackling both the computational limitations of the previous OBD implementations and the need to accommodate rate heterogeneity across lineages. They extend the analytical framework by accurately incorporating a broader range of fossil records, including those

without described and recorded morphological characters or reliable taxonomic information at the subclade level, which are often very numerous (Table 1). These developments allow us to tackle an important interrogation, namely assessing the relative contribution of phylogenies, various types of fossils, and rate heterogeneity to accurately reconstructing diversification histories. After developing and validating the OBD and OBDD models, we explore this question by applying a range of diversification models, accounting or not for different fossil data types and rate heterogeneity, to a cetacean dataset.

Table 1. Comparison of the diversification models discussed in this paper. Each column represents a model and each row indicates whether it includes a given feature. Coded fossils are those placed in the phylogeny; uncoded occurrences are fossil sampling times not placed in the phylogeny. Rate heterogeneity refers to branch-specific diversification rates evolving according to a geometric Brownian motion (GBM). Piecewise sampling rates allow fossil sampling rates to vary across user-defined time intervals.

	BD	FBD	OBD	BDD	FBDD	OBDD
Coded fossils	–	✓	✓	–	✓	✓
Uncoded occurrences	–	–	✓	–	–	✓
Rate heterogeneity (GBM)	–	–	–	✓	✓	✓
Piecewise sampling rates	–	✓	✓	–	✓	✓

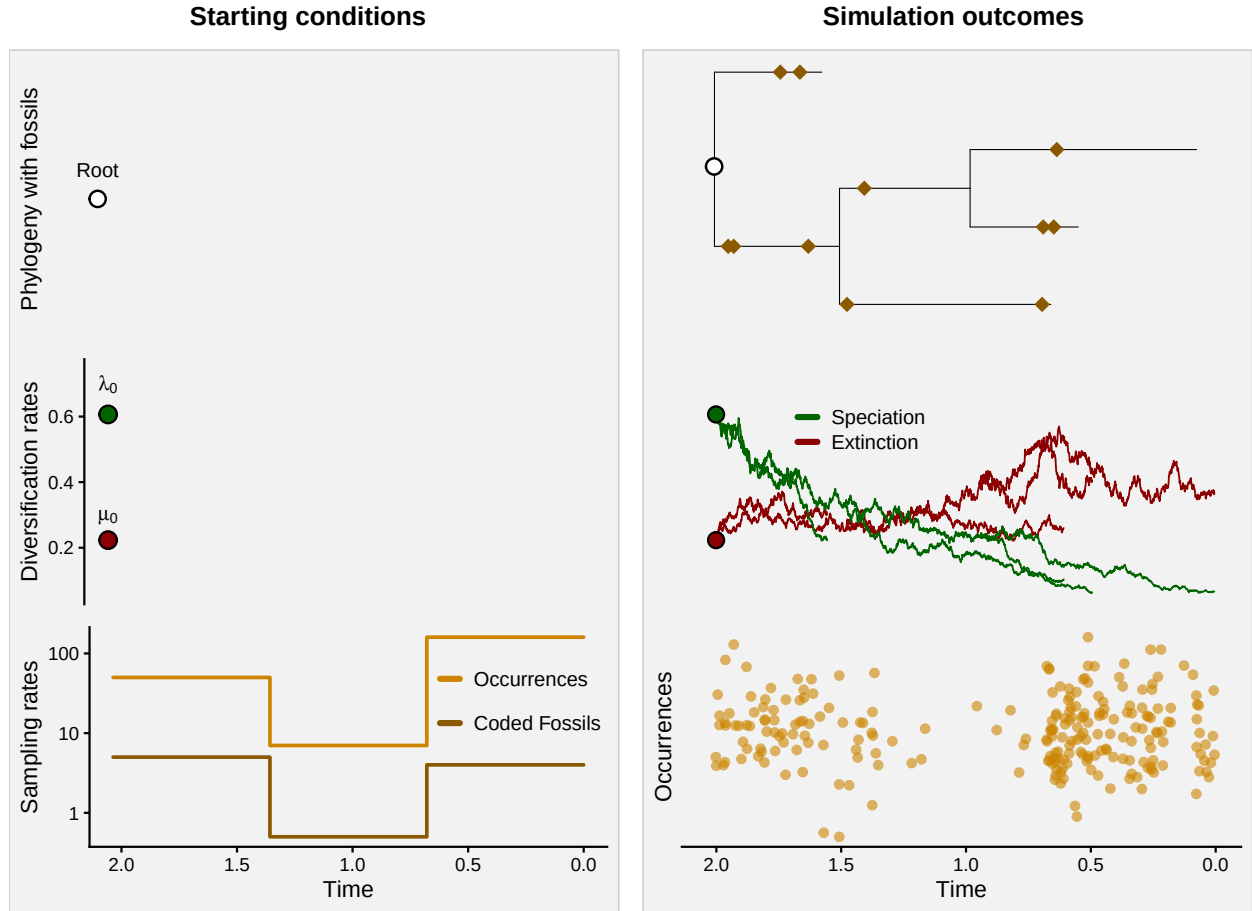


Figure 1. **The Occurrence Birth-Death Diffusion process.** This figure illustrates a simulation under the OBDD process. One key feature is that speciation and extinction rates are not specified in advance, except for starting values (resp. λ_0 and μ_0), but instead change gradually along each new branch of the phylogeny. The other important feature is that the sampling process produces **uncoded** occurrences that are not placed in the phylogeny, in addition to the standard **coded** fossils. Consequently, each simulation will result in both a sample of rate trajectories (right column, middle) and a phylogeny with fossils and a record of occurrences (right column, resp. brown diamonds and orange circles).

MATERIALS & METHODS

The Occurrence Birth-Death Diffusion process

We introduce the Occurrence Birth-Death Diffusion (OBDD) model, which combines a total-evidence approach of the fossil record with the flexibility of a branch-specific heterogeneous birth-death model (Fig. 1). To model heterogeneity in diversification rates

across the phylogeny, speciation and extinction rates are assumed to evolve according to a geometric Brownian motion (“GBM”), following Quintero et al. (2024). That is, the logarithms of speciation and extinction rates follow a Brownian motion (see Fig. 1) with variance parameters σ_λ^2 and σ_μ^2 , respectively. Furthermore, trend parameters α_λ and α_μ detect consistent rate increases or decreases over time. Specifically, the GBM processes start with initial values λ_0 and μ_0 , respectively, and follow:

$$\begin{cases} d \ln(\lambda_l(t)) = \alpha_\lambda dt + \sigma_\lambda W(t) \\ d \ln(\mu_l(t)) = \alpha_\mu dt + \sigma_\mu W(t), \end{cases} \quad (0.1)$$

with $\lambda_l(t)$, $\mu_l(t)$ being the branch-specific rates for lineage l at time t , and $W(t)$ the Wiener process (i.e., standard Brownian motion). At each speciation event, both daughter lineages inherit the current speciation and extinction rates of the parent lineage.

Coded fossils are incorporated into the phylogeny with a fossil sampling rate (ψ) following a Poisson point process along branches of the tree (Stadler, 2010; Heath et al., 2014). The integration of uncoded occurrences employs a similar approach, with a rate of occurrence sampling denoted by ω (Vaughan et al., 2019). The difference is that these occurrences are not included in the phylogeny but are instead only recorded as sampling times (as shown in Fig. 1). Note that this parametrization is equivalent to a model with a single total fossil sampling rate $\varphi = \psi + \omega$ and a probability $p = \psi / (\psi + \omega)$ that each fossil find is coded. The total number of finds follows a Poisson process with rate φ , and, conditional on the total, the number of coded fossils follows a Binomial distribution with probability p . The use of separate rates ψ and ω therefore does not imply different sampling regimes for coded and uncoded fossils, but rather allows variations in fossilization rates and coding probability to be estimated from the data. Given that fossilization and fossil recovery rates are expected to vary over geological time (Raup, 1975; Dominici et al., 2020), we allow these sampling rates to vary in a piecewise manner at user-defined times (Fig. 1).

Parameter inference with Bayesian data-augmentation

We implement the OBDD model within the computational framework of the Julia package Tapestry (Bezanson et al., 2017; Quintero and Andréoletti, 2020). Parameter inference uses a Bayesian data augmentation approach (Tanner and Wong, 1987), as described in prior studies on other phylogenetic diversification models (Maliot and Morlon, 2022; Barido-Sottani and Morlon, 2023) and on Birth-Death Diffusion processes in particular (Quintero et al., 2024, 2025). During a Markov Chain Monte Carlo (MCMC) procedure, the data augmentation algorithm involves alternately simulating missing parts of the phylogenetic tree, referred to as augmented branches, as well as all branch-specific diversification rates, and updating the model parameters based on this augmented data (Fig. 2). Therefore, a complete phylogeny, combining the reconstructed tree and the augmented branches, with associated diversification rates, is available at each iteration of the MCMC, simplifying the likelihood computation for any set of model parameters (see Appendix A). Data augmentation thus circumvents the difficulty of computing the likelihood of complex models based on the reconstructed phylogeny, which otherwise requires the analytical integration of unobserved components of the process. Furthermore, this iterative process generates a “forest” of posterior augmented trees, with associated diversification rates, which can be used to reconstruct the posterior distribution of species richness, or evolutionary rates, over time.

To model rate heterogeneity across the phylogeny, we discretise all branches at regular intervals, δt (set to 10^{-3} Myr, i.e. 1000 years, in all our analyses) and approximate the GBM described in the previous section. Gibbs sampling is used for most parameters, leveraging conjugate priors to enhance the efficiency of the inference process, and a Metropolis-Hastings algorithm is used for augmenting unobserved branches (Quintero et al., 2024, 2025). The sampling rate of coded fossils (ψ) is inferred similarly to the FBD and FBDD processes, using a conjugate Gamma prior updated based on the number of fossils and the total branch length of the complete augmented phylogenies, as in

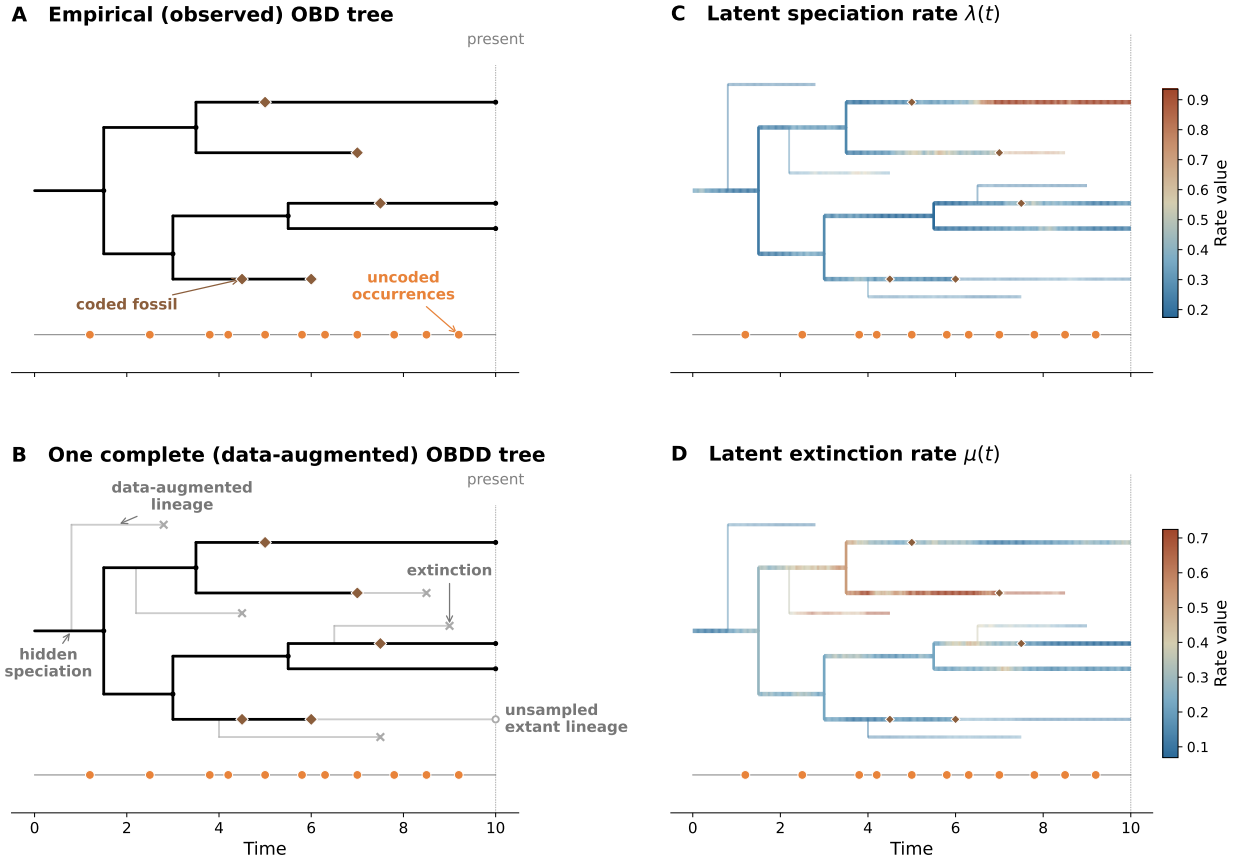


Figure 2. **The data augmentation approach in the OBDD model.** (A) The empirical (observed) tree includes coded fossils (brown diamonds) placed in the phylogeny, and uncoded occurrences (orange circles) recorded only as sampling times. (B) One complete data-augmented tree, as sampled during a single MCMC iteration. Gray lineages represent augmented branches, including hidden speciation and extinction events, as well as unsampled extant lineages. (C, D) Latent speciation (λ) and extinction (μ) rates along the augmented tree, where rates are inherited at speciation and then evolve independently along each lineage according to a geometric Brownian motion.

Quintero et al. (2025). In the following, we detail parts of the inference particular to the OBD and OBDD process, i.e., those involved in the integration of **uncoded** occurrences (sampling rate ω).

The MCMC procedure explores the parameter space guided by the likelihood of observed data for current and proposed parameter values. Given that we introduced **uncoded** occurrences, let us consider their likelihood according to our sampling process. On a single branch of length L_i with a constant sampling rate ω , the likelihood of

observing a specific number of **uncoded** occurrences k_i follows a Poisson distribution $\mathcal{P}(\omega L_i)$. When N_i lineages exist in parallel, the sum of all sampled occurrences also follows a Poisson process $\mathcal{P}(\omega L_i N_i)$. We apply this property across I intervals delimited by changes in the diversity-through-time curve of the augmented tree: each interval i has a constant number of lineages N_i of identical length L_i and contains k_i occurrences. This Poisson sampling assumption is shared with the TRiPS method (Starrfelt and Liow, 2016) and with the stratigraphic likelihood of Huelsenbeck and Rannala (1997). The probability of the occurrence data Ω given the complete augmented tree Ψ and the occurrence sampling rate ω is thus the following:

$$p(\Omega|\Psi, \omega) = \prod_{i=1}^I \frac{(\omega L_i N_i)^{k_i} e^{-\omega L_i N_i}}{k_i!}. \quad (0.2)$$

Let us compute the likelihood ratio for a new proposal parameter value ω' relative to the current value ω :

$$\begin{aligned} \frac{p(\Omega|\Psi, \omega')}{p(\Omega|\Psi, \omega)} &= \prod_{i=1}^I \left(\frac{\omega'}{\omega} \right)^{k_i} e^{-(\omega' - \omega) L_i N_i} \\ &= \left(\frac{\omega'}{\omega} \right)^k e^{-(\omega' - \omega) L} \end{aligned}$$

where $k = \sum_{i=1}^I k_i$ is the total number of occurrences and $L = \sum_{i=1}^I L_i N_i$ is the cumulative length of all observed and augmented lineages. Note that the exact timing of the occurrences does not affect the likelihood ratio; the process is equivalent to considering a single Poisson process $\mathcal{P}(\omega L)$ on the full augmented tree. With piecewise-constant sampling rates, the same reasoning applies within each time period. Thus, the posterior distribution of ω_p in a period p can be efficiently sampled using a conjugate Gamma prior, $\Gamma(\gamma_1, \gamma_2)$, which is updated based on the observed **uncoded**

occurrences k_p and sum of branch lengths L_p during this period:

$$p(\omega_p|\cdot) = \Gamma(\gamma_1 + k_p, \gamma_2 + L_p).$$

The Gamma distribution is a natural choice of prior for the sampling rates ψ and ω per time since the waiting times are exponentially distributed. This results in conjugacy with the likelihood of Poisson processes, enabling closed-form Gibbs updates and more efficient MCMC behaviour with faster mixing and increased effective sample size. Note that these rates per unit time are different from “occupancy probability” (the proportion of sampling units such as sites or collections in which a taxa is found) (Foote, 2016) for which previous empirical studies have estimated a LogNormal distribution, conditioned on a known number of collections (Wagner and Marcot, 2013; Foote, 2016).

The data augmentation steps involve proposing modifications to the tree, that is, replacing some of the reconstructed branches by subtrees simulated under the process, and accepting or rejecting these proposals based on Hasting ratios. Here, the acceptance ratio used for the inference of the FBDD process (Quintero et al., 2025) is modified to account for the likelihood ratio of the occurrence data given the proposed and current augmented phylogenies. The likelihood of observing the **uncoded** occurrences given an augmented tree depends only on the number of lineages over time, represented by the Diversity-Through-Time (DTT) trajectory. Modifying the augmented tree alters the DTT, which in turn affects the likelihood of the **uncoded** occurrence data. Using the likelihood expression in Equation 0.2 over J intervals delimited by changes in the current or proposed DTT, we derive the following likelihood ratio of the proposed tree Ψ' compared to the current tree Ψ :

$$\begin{aligned}
\frac{p(\Psi', \Omega | \cdot)}{p(\Psi, \Omega | \cdot)} &= \frac{p(\Psi' | \cdot)}{p(\Psi | \cdot)} \frac{p(\Omega | \Psi', \omega)}{p(\Omega | \Psi, \omega)} = \frac{p(\Psi' | \cdot)}{p(\Psi | \cdot)} \times \prod_{j=1}^J \left(\frac{\omega L_j N_j^{\Psi'}}{\omega L_j N_j^{\Psi}} \right)^{k_j} e^{-\omega(L_j N_j^{\Psi'} - L_j N_j^{\Psi})} \\
&= \frac{p(\Psi' | \cdot)}{p(\Psi | \cdot)} \times \prod_{j=1}^J \left(\frac{N_j^{\Psi'}}{N_j^{\Psi}} \right)^{k_j} \times e^{-\omega \sum_{j=1}^J L_j (N_j^{\Psi'} - N_j^{\Psi})}.
\end{aligned}$$

The first part of this ratio is identical to the one used in the FBDD process (see Appendix A), while the added terms ensure that at each iteration the augmented trees reflect the observed **uncoded** occurrence record. In other terms, the greater the number of **uncoded** occurrences included in a given period, the more likely it is that many augmented branches will be accepted in that interval, which will then influence the diversification rate estimates.

The simpler OBD model with constant homogeneous speciation and extinction rates is also implemented in Tapestry.jl (Quintero and Andréoletti, 2020) but as an independent model, extending the constant-rate FBD model (Quintero et al., 2025). The integration of **uncoded** occurrences relies on the same functions for the Gibbs sampling and Metropolis-Hastings steps described above.

Simulation-based model validation

We validate both the OBD and OBDD models using a procedure called simulation-based calibration, which consists, for a given model, of:

1. choosing prior distributions for the model parameters;
2. simulating many trees using parameter values drawn from these distributions (here, we simulated 1000 trees under the OBD model, and 500 under the OBDD model);
3. for each simulated tree, inferring posterior distributions for all parameters, starting from the prior distributions above;

4. counting the recovery proportion of the true parameter values in posterior credible intervals of increasing widths (from 5% to 95%).

If the recovery proportion matches the widths of all credible intervals, the model is considered well calibrated.

For the OBD model, the initial speciation and extinction rates were drawn from a Gamma(1.0, 1.0) distribution, with a time of origin set to 2.0 units. For the OBDD model, the initial speciation and extinction rates were drawn from Gamma(1.5, 1.0) distributions, trend parameters α_λ and α_μ from Normal(0.0, 0.1) distributions, and diffusion parameters σ_λ and σ_μ from InverseGamma(5.0, 0.5) distributions. The time of origin was set to 1.0 units. These distributions were selected to produce phylogenies of moderate size for computational efficiency, while covering realistic parameter values (Quintero et al., 2025). Coded fossil sampling rates ψ and uncoded occurrence rates ω were modelled with piecewise constant rates over two epochs, divided at half the time of origin, and are now denoted as $\psi_1, \psi_2, \omega_1, \omega_2$. They were drawn, respectively, from the following distributions: $\psi_1, \psi_2 \sim \text{Gamma}(1.0, 1.0)$ and $\omega_1, \omega_2 \sim \text{Gamma}(1.0, 2.0)$ distributions for OBD and $\psi_1, \psi_2 \sim \text{Gamma}(2.0, 1.0)$ and $\omega_1, \omega_2 \sim \text{Gamma}(2.0, 5.0)$ for OBDD, reproducing the expected higher number of uncoded occurrences compared to coded fossils in the tree. MCMC inferences were run for 500.000 and 2.000.000 iterations, respectively. We recorded the total running time of each MCMC inference, in order to analyse how runtime scales with phylogeny size and the inclusion of fossils.

Case study: Cetacea

We used the Cetaceans as a case study to explore the effect on the inference of diversification histories of i) the inclusion of fossils, in the phylogeny or as simple occurrences, and ii) accounting for rate heterogeneity. Cetaceans are well suited for this purpose because they have one of the most extensively curated fossil records in the Paleobiology Database, with broad taxonomic and temporal coverage. We used the

cetacean phylogeny from Lloyd and Slater (2021), which contains 89 extant species and 405 coded fossil taxa (each representing a distinct morphotaxon). This phylogeny was reconstructed through a meta-analytic approach to combine previous datasets into a metatree that was then used to set constraints on a timetree inference under an FBD tree prior. More specifically, we used five phylogenies drawn from the set of posterior phylogenies provided in the paper (identified as the “Safe” posterior) as replicates for each analysis. All diversification analyses presented here operate on these fixed phylogenies; only the diversification rates and augmented (unobserved) branches are inferred. In addition, we used the 966 uncoded occurrences from the Paleobiology Database (Uhen et al., 2023; McClennen, 2024) used in Andréoletti et al. (2022), and we sampled their age uniformly in their age uncertainty range at the beginning of each replicate. Their temporal distributions over the five replicates are presented in Appendix C.

We fitted eight diversification models, all implemented in Tapestry.jl:

- the constant-rate Birth-Death, Fossilized Birth-Death, Occurrence Birth-Death, and Occurrence Birth-Death without coded fossils models (BD, FBD, OBD, and OBD-noF);
- the Birth-Death Diffusion, Fossilized Birth-Death Diffusion, Occurrence Birth-Death Diffusion, and Occurrence Birth-Death Diffusion without coded fossils models (BDD, FBDD, OBDD, and OBDD-noF).

For the BD and BDD models, we removed coded fossils from the phylogenies, as well as the stem branch, whose estimated length might not be available in the absence of fossils. For the OBD and OBDD models, we added uncoded occurrences. For the OBD-noF and OBDD-noF models, we removed fossils from the phylogeny but included uncoded occurrences to assess the relative contribution of coded fossils and uncoded occurrences. We kept the stem branch, assuming that its length was estimated if occurrences were available. For models including fossilization (i.e., all models except the

BD and BDD models), we accounted for known differences in fossil availability across geological periods by considering piecewise constant temporal variations in fossilization rates (ψ for [coded](#) fossils and ω for [uncoded](#) occurrences), enabling rate changes at the start and end of the [Rupelian \(Early Oligocene\)](#), [Aquitania \(Early Miocene\)](#), and [Messinian \(Late Miocene\)](#) periods of well-established low preservation (Marx et al., 2016; Dominici et al., 2020), as in Andréoletti et al. (2022). We thus inferred seven [coded](#) fossil sampling rates and seven [uncoded](#) occurrence sampling rates for the OBDD model. Table 2 details the data, the number of parameters, and the MCMC inference efficiency of each of these models on the cetacean dataset. [The cetacean dataset required long runtimes because cetaceans experienced a pronounced boom-and-bust diversification history, with a large radiation followed by widespread extinction. Consequently, many past lineages left no extant descendants and were not sampled as fossils, requiring the MCMC to simulate a large number of augmented branches to represent these unobserved lineages.](#) MCMC runs that were still ongoing after 100 days were interrupted. Convergence was evaluated using Tracer (Rambaut et al., 2018) and the R library *coda* (Plummer et al., 2006), with the average Effective Sample Sizes (ESS) reported in Table 2. This evaluation focuses on parameter convergence rather than augmented tree convergence, as current phylogenetic methods for assessing phylogenetic convergence are limited to trees with known tips. Nevertheless, the two should be closely related in the data augmentation procedure.

Table 2. Summary of the different diversification models and corresponding analyses on the cetacean dataset. The table shows for each analysis the number of model parameters, characteristics of the cetacean data subset included in the inference - number of tips, **coded** fossils and **uncoded** occurrences - and the computational efficiency. Models include homogeneous variants (BD, FBD, OBD, OBD-noF) and diffusion-based variants (BDD, FBDD, OBDD, OBDD-noF) ; the suffix -noF indicates models run without **coded** fossils in the phylogeny, retaining only **uncoded** occurrences. MCMC durations are averaged over 5 replicates. Effective sample sizes (ESS) are averaged over replicates and parameters.

Model	BD	FBD	OBD	OBD-noF	BDD	FBDD	OBDD	OBDD-noF
Parameters	2	9	16	9	5	13	20	13
Tips	89	372	372	89	89	372	372	89
Coded fossils	0	405	405	0	0	405	405	0
Occurrences	0	0	966	966	0	0	966	966
Burn-in	50K	100K	100K	100K	50K	50K	50K	50K
Iterations	2M	2M	2M	50M	10M	10M	10M	20M
Duration (days)	0.01	0.07	0.85	2.9	7.7	100	100	80
Average ESS	1300	1800	1400	740	810	1900	1800	310

RESULTS

Validating the OBD and OBDD implementations

Our simulation-based calibration analyses of the Occurrence Birth-Death (OBD) and Occurrence Birth-Death Diffusion (OBDD) models demonstrate a well-calibrated implementation for all parameters (see Appendix D and Fig. 3). The recovery proportions of the true parameter values consistently match the credible interval widths (Fig. D.1a and 3a), and the estimation errors decrease with increasing tree size (Fig. D.1b and 3b). Overall, the validation confirms that both models perform reliably in recovering true diversification parameters from simulated data. Although we did not directly compare simulated and inferred branch-specific diversification rates, the proper recovery of the model hyperparameters implies that the branch-specific rates are also well recovered. Many of the simulated phylogenies are of moderate size, which is enough to validate the correctness of our inference algorithm with the calibration study,

and generalises to larger phylogenies.

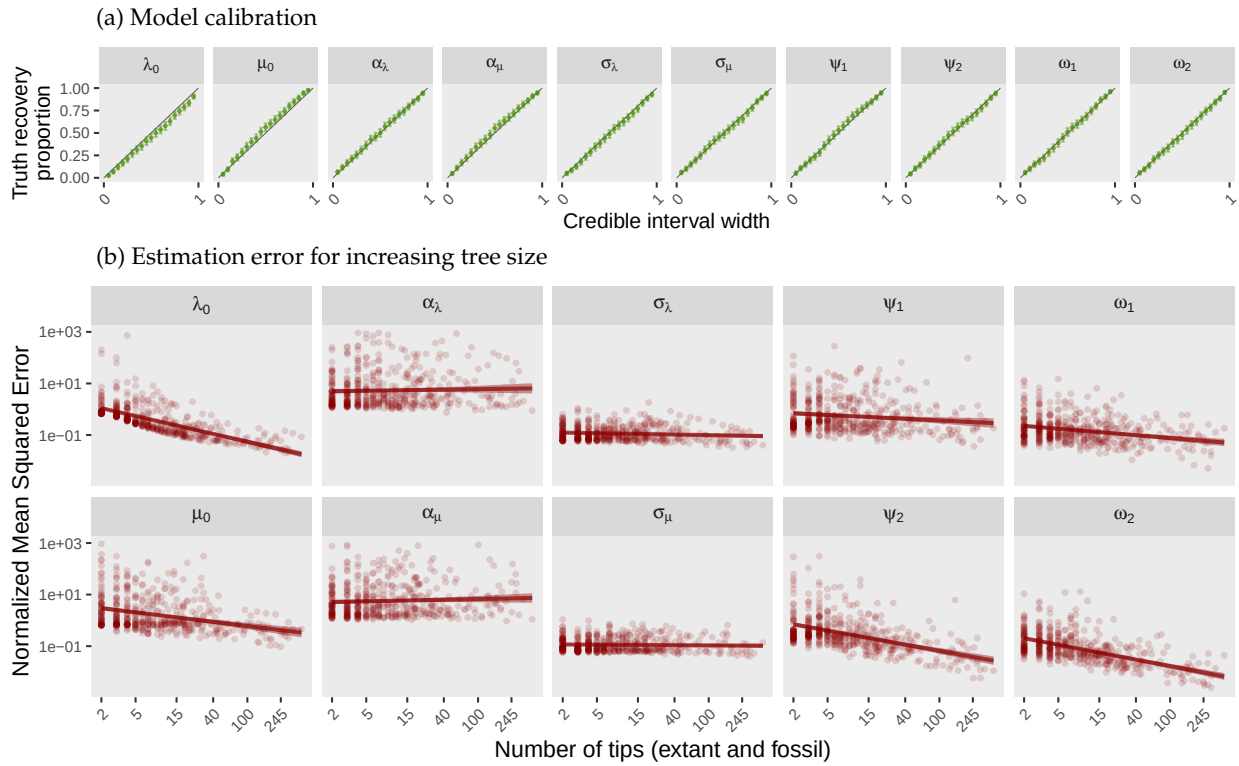


Figure 3. **Simulation-based validation of the OBDD model.** Results of the simulation-based calibration study on 500 datasets simulated under the OBDD model. (a) Truth recovery proportions: the proportion of datasets for which the true parameter value falls within posterior credible intervals of increasing width (from 5% to 95%). A well-calibrated model produces points along the diagonal. (b) Estimation error, measured as the normalized mean squared error NMSE $(\hat{\theta} - \theta)^2 / \theta^2$ (where $\hat{\theta}$ is the posterior median and θ the true value), plotted against the number of tips in the reconstructed phylogeny.

Regarding the influence of tree size on computation time, Figure 4 shows that, for datasets generated under the OBDD model and for a fixed number of MCMC iterations, our inference should take an extra day for each additional 150 tips in the reconstructed phylogeny (including fossil tips). This estimate is much lower than the runs on the cetacean dataset, because the strong fluctuations in diversification rates the cetaceans exhibit are unlikely under the geometric Brownian motion (GBM) prior used to generate the calibration datasets, which produces relatively smooth, gradual rate variation. The

calibration datasets therefore contain fewer unobserved lineages, and correspondingly fewer augmented branches to simulate leading to shorter per-iteration runtimes. For some empirical cases where computation time might become impractical, one possible partial solution is to take advantage of the fast convergence under our Gibbs sampling algorithm and run parallel chains to be later aggregated to achieve higher Effective Sample Size. The Generalised Linear Model (GLM) and Generalised Additive Model (GAM) regressions (see Appendix B) revealed no influence of the number of fossils or occurrences on the runtime, as apparent in Figure 4, indicating that the code has been adequately optimised for extensive fossil integration.

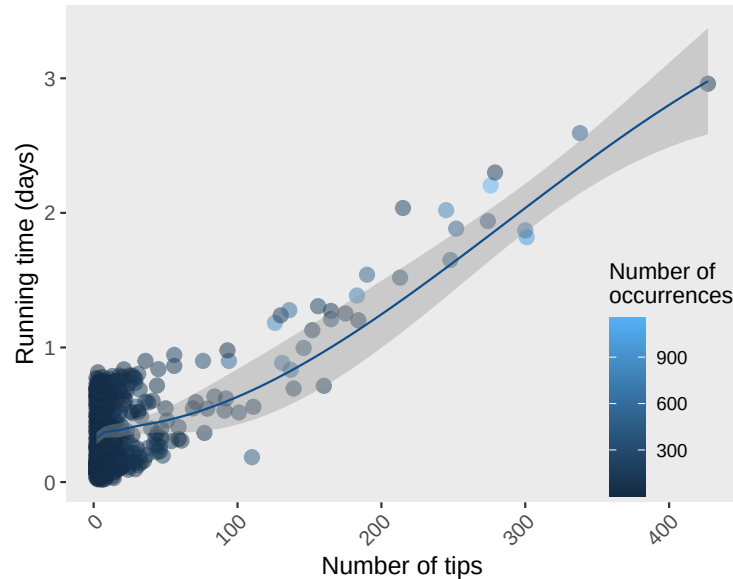


Figure 4. **Scaling of MCMC runtime with tree size for the OBDD model.** Each point corresponds to one of the datasets used for the simulation-based model validation procedure, from which we recorded the number of tips in the reconstructed phylogeny (including fossil tips) and the runtime for the set number of iterations. The colour gradient indicates the number of simulated **uncoded** occurrences. **Note that these runtimes reflect an ideal scenario in which datasets are generated directly under the model, producing relatively smooth diversification dynamics and fewer augmented branches to simulate. Empirical datasets with more complex histories (e.g. cetaceans) require substantially longer runtimes; see Table 2 for empirical values.**

In Appendix E, **using the same simulated datasets**, we further explore the drivers of estimation error for each parameter, **and investigate potential estimation biases by**

reporting absolute errors. Absolute errors confirm the absence of systematic bias. We observe that the NMSE error reduction for several parameters is mostly driven by their corresponding data: extant tree size for the initial speciation and extinction rate, number of fossils for the fossil sampling rates, and number of occurrences for the occurrence sampling rate. The drift and diffusion parameters are generally more difficult to infer (see Fig. 3b), and no significant predictor influence was detected. The interaction between fossils ($\sqrt{n_{\text{fossils}}}$) and occurrences ($\sqrt{n_{\text{occ}}}$) also has an influence on the estimation error, which was positive for most parameters, although only significant for ψ_2 and ω_2 . The positive effect reflects that as the number of fossils included in the phylogeny increases, the information provided by additional occurrences decreases, and vice versa.

Assessing the role of fossils in estimating the diversification history of Cetaceans

Consistent with previous results on mammals (Quintero et al., 2024), we found that models without fossils infer overly simplistic species richness trajectories, close to the observed lineages through time (LTTs) of the reconstructed phylogenies (Fig. 5a and 5e). These models fail to capture the rise and fall trajectory of cetacean diversity expected from paleontological records over the Cenozoic (Morlon et al., 2011; Marx and Fordyce, 2015; Quintero et al., 2025). In models that include only uncoded occurrences but exclude coded fossils (OBD-noF, OBDD-noF, Fig. 5b and 5f), we recover diversity estimates of similar magnitude to the inferred richness trajectories by models with fossils fully integrated into the phylogeny (Fig. 5c–d and 5g–h). However, the uncertainty of the estimate is much higher.

The diffusion models (BDD, OBDD-noF, FBDD, OBDD) infer richness trajectories that are qualitatively similar to their homogeneous counterparts, albeit with higher maximum richness levels (especially for OBDD-noF). However, looking at the average diversification rates over time, Fig. 6g and 6h demonstrate that, even though the overall richness patterns are comparable, the incorporation of rate diffusion detects clear

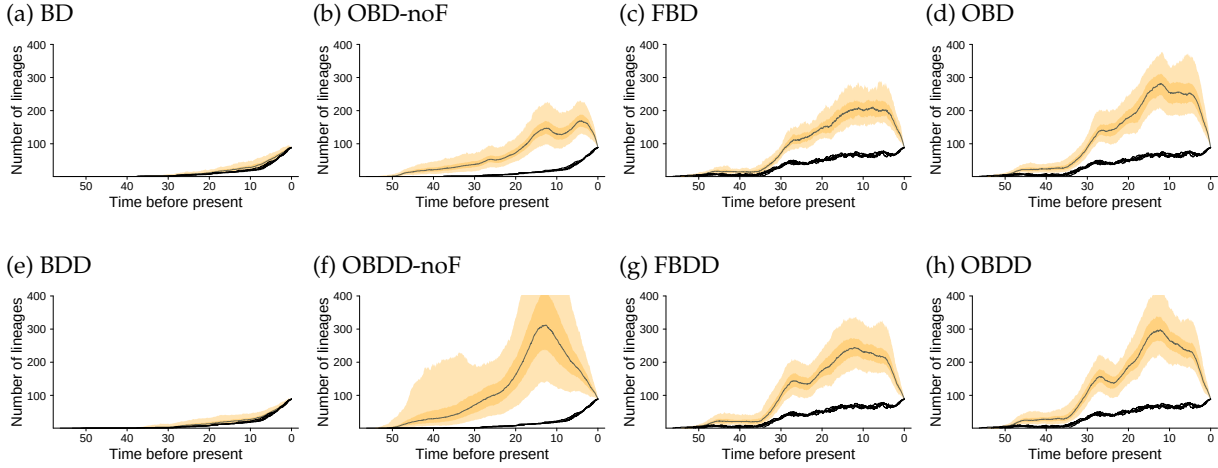


Figure 5. Inferred species richness over time across models. Each panels shows the inferred number of cetacean lineages over time with a given model, conditioned on the reconstructed tree and recorded **uncoded** occurrences. The gray line is the median value across the augmented trees in the combined MCMC samples from the 5 replicate analyses. The 5 overlapping black lines represent the lineage-through-time curves from the reconstructed phylogeny of each replicate. Dark and light yellow areas represent 50% and 95% credible intervals.

variations in the rate trajectories. Both the FBDD and OBDD models infer a rise of speciation rates up to 35 million years ago, followed by a slow decrease, while extinction rates exhibit three major peaks at approximately 40, 27, and 3 Ma, either matching or most recently exceeding speciation rates. In contrast, the BDD model, as shown in Figure 6e, infers low and nearly constant speciation rates (~ 0.15 per million species years, MSY), with extinction rates that steadily decline over time and are lower than speciation rates for most of the group's history. Constant rate models infer speciation and extinction rates that are close to averaged heterogeneous rates (see panels 6a to 6d). The BD model (6a) infers the lowest rates and higher speciation than extinction, while the models with **coded** fossils and/or occurrences infer negligible net-diversification rates.

The reconstructed diversity trajectories for constant-rate models with fossils (Fig. 5b–d) may seem surprising given the estimated diversification rates (Fig. 6b–d), as hump-shaped diversity curves are not expected under constant net diversification rates. In fact, the diversity curves reconstructed here should not be interpreted as expected

diversity curves under the inferred models, as in Etienne et al. (2011); Billaud et al. (2019), but as curves conditioned on the observed data. The past diversity estimates are indeed reconstructed from the distribution of augmented complete phylogenies produced by the data augmentation procedure, which is conditioned on the reconstructed phylogeny and **uncoded** occurrences. This distribution is therefore equivalent to the diversity curve conditioned on the data for the homogeneous OBD model described in Manceau et al. (2021), whose algorithm was implemented by Andréoletti et al. (2022) in RevBayes.

The remaining panels in Figure 6 show sampling rates for the models with fossils. **Uncoded** occurrence sampling rates are higher than those of **coded** fossils, as expected, and the inferred rates in each time period are generally very similar between homogeneous (6j–i) and heterogeneous (6m–l) models. Both sampling rates are reduced in the 3 delimited geological periods of expected low preservation (see Methods), and particularly in the Rupelian (33.9–27.3 Ma).

Heterogeneous Cetacean diversification

The inclusion of heterogeneous rates in diffusion models reveals significant variation in diversification rates between different clades. Based on the phylogeny of extant species, the BDD model infers markedly higher net-diversification rates for the Delphinidae family, up to 0.4 per Million Species Years (MSY), as shown in Figure 7m, compared to other lineages that exhibit rates between -0.1 and 0.2 per MSY. This difference is mainly due to higher speciation rates in this family (panel 7e) rather than reduced extinction.

The diffusion models with **coded** fossils, including FBDD and OBDD, capture even stronger levels of rate heterogeneity across cetacean lineages (panels 7o and 7p), with most clades exhibiting high net-diversification rates at their origins, followed by a progressive decline, except for those that gave rise to modern cetacean families. This trend is due, in most individual lineages, both to a strong decline in speciation rate (panels 7f–h) and to an increase in extinction (panels 7j–l).

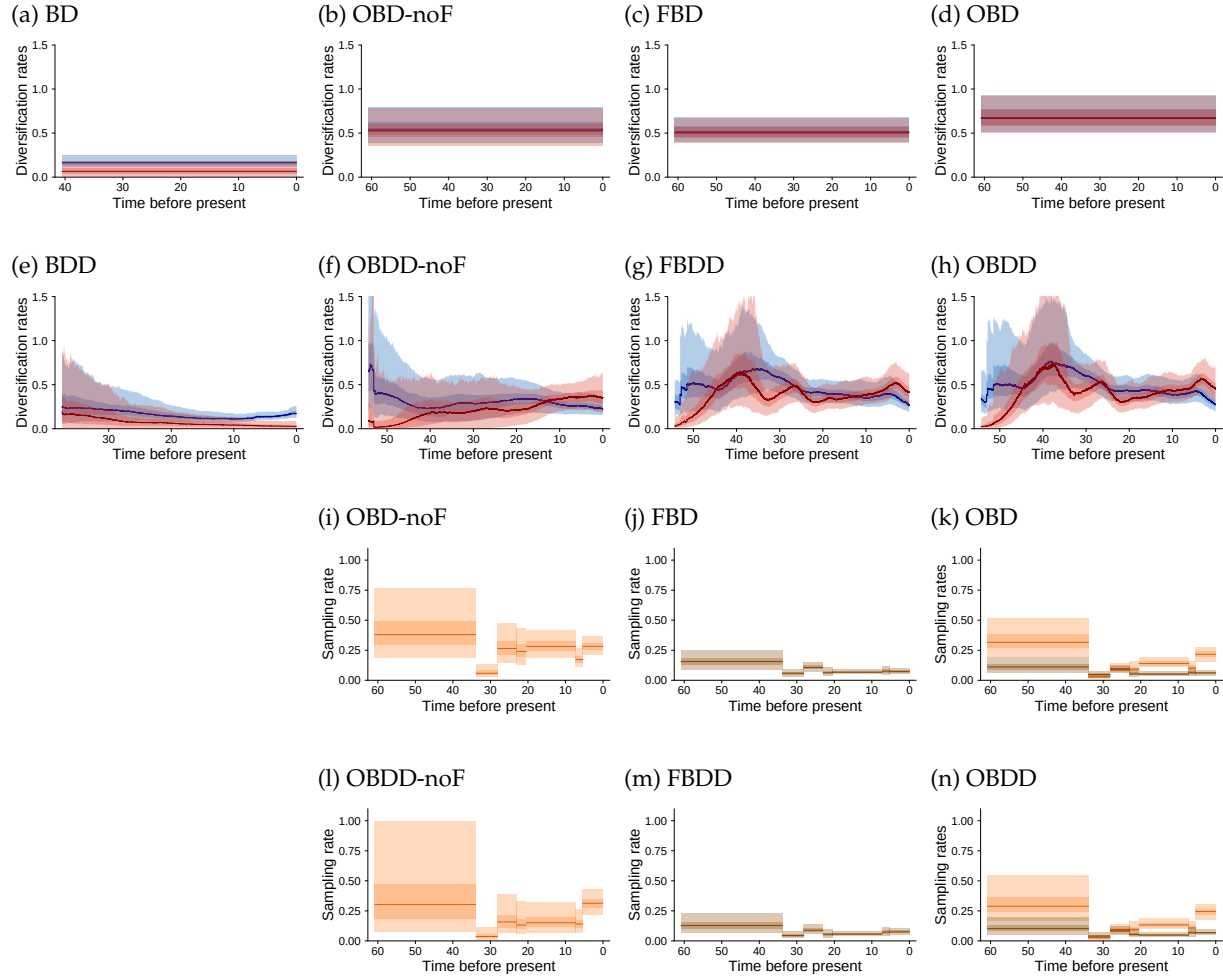


Figure 6. Inferred cetacean diversification and sampling rates across diversification models. Panels a-h show inferred speciation (blue) and extinction (red) rates over time. The line is the median value averaged at each given time over all the branches, observed or augmented, in the combined MCMC samples of the 5 replicate analyses. Panels i-n show inferred piecewise-constant sampling rates of coded fossils (brown) and uncoded occurrences (orange) over time, when applicable. For all panels, dark and light shaded areas represent 50% and 95% credible intervals.

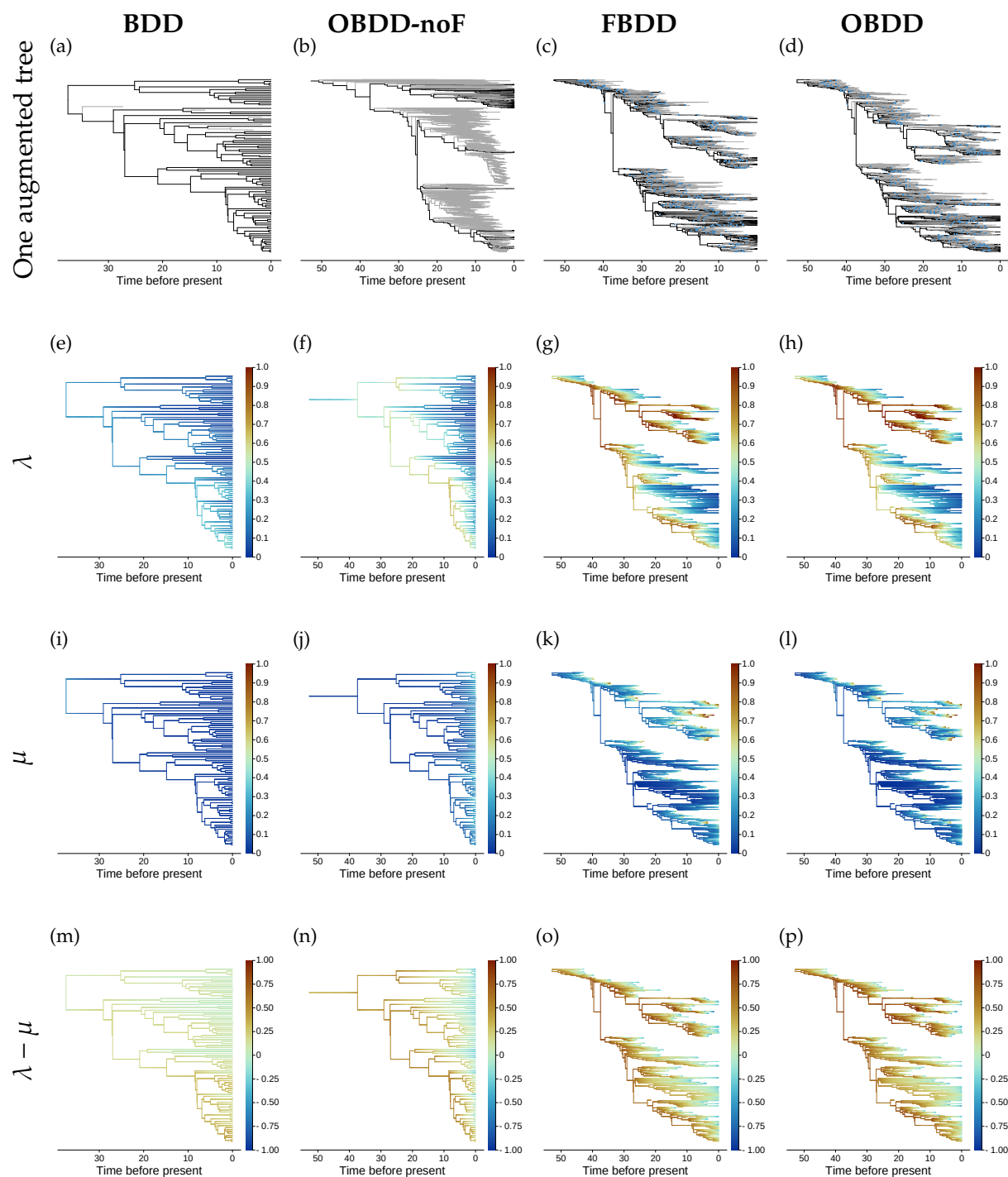


Figure 7. **Branch-specific cetacean diversification rates across diffusion models.** Panels a-d display samples of complete phylogenies from the MCMC posteriors, with reconstructed branches in black and augmented branches in gray. The remaining panels show the median posterior rates on the reconstructed phylogeny, with branch-specific speciation, extinction and net-diversification rates on the second (e-h), third (i-l) and fourth (m-p) row, respectively.

DISCUSSION

Fossils are key to understand the past under BDD models

Our results demonstrate the importance of incorporating fossil evidence to accurately reconstruct diversification histories. The two models we used here that rely solely on phylogenies (BD and BDD) drastically underestimate the historical diversity of cetaceans reflected in the fossil record. Since these models are well calibrated, we assume that the discrepancy arises from model misspecification, specifically imposing constant and homogeneous diversification rates for the BD model and geometric Brownian motion dynamics for the BDD model. The integration of fossil evidence likely mitigates the impact of these misspecifications. Morlon et al. (2011); Mazet et al. (2023) were able to recover a humped-shaped diversity pattern from a phylogeny of extant cetacean species, suggesting that the model they used, which allowed large shifts in speciation and extinction dynamics at the base of major families, better represents the true dynamics of cetaceans. However, making such a priori hypotheses could in turn be the source of model misspecification in other clades. Instead, the flexibility offered by models such as the BDD, if complemented with fossil information, can recover accurate dynamics and be more immune to misspecification issues. In the case of cetaceans, BDD models enhanced with fossil data were able to recover diversification shifts at the base of families and boom-bust diversity dynamics, even though such dynamics are rather unlikely under the BDD process. This is consistent with the findings of Quintero et al. (2024) showing that BDD models can recover clade-wide shifts in diversification rates.

The integration of **uncoded** occurrences shows the value of these additional data, in particular in the absence of fossils whose phylogenetic placement can be inferred: **uncoded** occurrences alone improve the reconstruction of hidden diversity. However, they are less informative than **coded** fossils placed in the phylogeny, especially when many of these fossils are available, as is the case for cetaceans. We expect occurrences to be most valuable for accurately estimating diversification rates in epochs where fossil

information from phylogenies is sparse, or in groups that lack such fossils altogether. In some groups, it should be possible to use other data reflecting a clade's total species richness over time as a type of occurrence that can be used in addition to the phylogeny. These alternative occurrences could include isolated teeth or scales (e.g., in sharks or other non-mammalian vertebrates, where such remains are abundant but often lack the diagnostic value needed for phylogenetic placement), specific biochemical proxies of the studied group, plant organs (leaves, seeds, pollen), or trace fossils such as coprolites. Many of these remains can be identified as part of the clade, but cannot be associated to other parts of the whole organism that are used for species identification (e.g. the skeleton) and are as such described in parallel parataxonomies that usually cannot be mapped to the clade's main taxonomy. This fact thus prevents them from being incorporated in the phylogeny in an FBD analysis, but not as uncoded occurrences in an OBD analysis, as long as they can be unambiguously attributed to the clade of interest. We note that for well-studied groups such as cetaceans, where taxonomic effort has been relatively uniform across geological periods, the probability that a given fossil is coded (as opposed to remaining an uncoded occurrence) is likely approximately constant over time. The OBDD model does not impose this constraint, allowing the coding probability to vary across user-defined time periods, which may be relevant for clades with less extensive taxonomic attention.

Importantly, the fossil record is inherently incomplete, limiting our direct observation of past diversity. Several non-phylogenetic approaches also account for unsampled lineages when estimating richness or diversification rates using rarefaction, Poisson processes, lower bound extrapolation or neural networks (Alroy, 2008; Foote, 2001; Starrfelt and Liow, 2016; Flannery-Sutherland et al., 2022; Cooper et al., 2024). The phylogenetic modeling framework, including all models presented here, complements these approaches by using phylogenetic relationships to inform the placement and timing of unsampled lineages. Ghost lineages (branches implied by the tree topology

between divergence points) were accounted for by the phylogenetic reconstruction itself, before running our analyses. Then, our data augmentation procedure simulates Signor-Lipps extensions (extending known species ranges forward in time) and entirely unsampled lineages that originate, diversify, and go extinct without leaving any observed trace as in fossilised birth-death models (see Fig. 2). Other phylogenetic diversification models integrate over them analytically (Stadler, 2010). By jointly modeling the process of speciation, extinction and sampling, the FBD model provides a mathematical coherent framework to estimate unobserved diversity that accounts for the interaction between sampling, speciation and extinction (such as a higher probability of sampling for clades undergoing faster diversification).

Incorporating rate heterogeneity in phylogenetic models refines our view of a clade's evolution

Interestingly, rate heterogeneity appears to have a limited influence on the estimated past diversity, which is already well constrained by `coded` fossils and `uncoded` occurrences. This is notable given that heterogeneous models were compared to constant-rate homogeneous models, whereas time-varying models may capture additional temporal dynamics. The inclusion of heterogeneity in the diffusion models (BDD, FBDD, OBDD) is thus most valuable for flexibly capturing variations in diversification rates at various scales, and using them to generate hypotheses on the evolutionary pressures, such as environmental changes or clade-specific factors, that may have influenced the diversification of a clade. These hypotheses can then be tested using models specifically designed for this purpose (Morlon et al., 2024).

At the individual lineage level, we observe a general decline in net-diversification rates (similar to the “insidious loss of species macroevolutionary fitness” described in Quintero et al. (2025)), except for a few lineages that maintain their speciation potential and give rise to most of the subsequent diversity. This selection effect matches the pattern of imbalanced speciation pulses observed in Quintero et al. (2024), as well as the

theoretical predictions under a model of evolutionary tempo by Budd and Mann (2025).

The diversification history of cetaceans

The occurrence-based models (OBD and OBDD) recover a hump-shaped pattern, in line with previous fossil and molecular studies (Morlon et al., 2011), but with higher diversity. In particular, they reproduce the rapid decline in the past 5 Myr that has previously been linked to global climate cooling and the Northern Hemisphere glaciation (Andréoletti et al., 2022). This pattern broadly matches the one obtained at the genus level in Andréoletti et al. (2022), although it differs in including a short decline in the late Oligocene (25 Ma) and a maximum attained in the Mid-Miocene (12 Ma) instead of the Pliocene (around 4.5 Ma).

Looking at the average rate trajectories over time, especially the ones inferred with the FBDD and OBDD models which incorporate the most information, it appears that extinction is the main driver of variations in diversification trends on the scales of millions of years, although it is concomitant with a long-term decline in speciation rates that together explain the decline of cetaceans. The second spike of extinction around 27 million years ago corresponds to the extinction of many archaeocetes, the stem cetacean lineages, soon after the emergence of the two modern groups, Mysticeti and Odontoceti. These diversification rate trajectories are consistent with the trends inferred by Quintero et al. (2025) using a variant of the FBDD model. Their analysis incorporates additional **uncoded** occurrences to inform the **coded** fossil sampling rate in order to correct for the common failure to include multiple fossils of the same morphospecies, but this only includes **uncoded** occurrences of morphospecies present in the phylogeny. Despite these similar trends, our estimated rates are approximately 50% higher overall than those reported by Quintero et al. (2025), and the lineage diversity in our FBDD and OBDD results is nearly double. This discrepancy likely arises from our approach of **separately estimating the sampling rates for coded fossils and uncoded occurrences (which, as**

noted in Methods, is equivalent to estimating a total sampling rate and a coding probability). In Quintero et al. (2025), including uncoded occurrences as additional sampled ancestors increases the sampling rate in the tree, which in turn decreases the estimated diversity, speciation rates, and extinction rates.

Considering variations among cetacean lineages, most long-lived species tend to see their net-diversification rate progressively decline, including many modern species. An exception is the Delphinidae family, which diversified rapidly in the past 10 Myr and displays higher net diversification rates close to the present. This likely reflects a combination of late Miocene–Pleistocene oceanographic reorganizations (closure of Tethys and the Central American Seaway, fluctuating seaway connections around southern Africa) and large sea-level oscillations, which repeatedly created opportunities for allopatric splits across oceanic basins, together with some delphinid traits (broad diets, social foraging) that allowed them to thrive under shifting, nutrient-rich environments driven by upwelling (Gowans et al., 2007; Steeman et al., 2009; do Amaral et al., 2018). The findings align with Barido-Sottani and Morlon (2023), which identified the highest speciation rates in the Delphinidae family using ClaDS, another heterogeneous diversification model with branch-specific rates. A key methodological difference between the two studies is that Barido-Sottani and Morlon (2023) inferred the phylogeny together with the diversification rates, while we applied here the diffusion models on a fixed reconstructed phylogeny. This phylogeny was obtained using an FBD prior with homogeneous diversification rates, which could generate biases (Barido-Sottani and Morlon, 2023). The convergence of results obtained here versus in Barido-Sottani and Morlon (2023) suggests that the sequential approach (i.e., reconstructing the phylogeny first) can be robust. Still, it would be useful in the future to allow joint reconstruction inference with diffusion models.

CONCLUSION

We introduced the Occurrence Birth-Death Diffusion (OBDD) process, a new model that integrates **uncoded** occurrences and accommodates heterogeneous diversification rates across lineages. Our analyses show that fossils are essential for accurately reconstructing diversification histories under diffusion models. **Uncoded** occurrences provide additional refinements that are particularly useful when **coded** fossils are sparse. In cetaceans, accounting for rate heterogeneity revealed important variations among lineages and during lineage lifetimes, as well as several periods of increased extinction.

From a methodological standpoint, the data augmentation approach used here simplifies the likelihood formula by conditioning on complete (augmented) trees, making it straightforward to extend the model, for instance to incorporate rate heterogeneity or piecewise sampling, without deriving the likelihood of the (unaugmented) tree analytically. There is currently no analytical solution for the latter under the BDD model, for example. In some cases, this flexibility comes at the cost of an expanded state space that the MCMC must explore, so data augmentation is best suited to settings where the augmented data remain tractable, when there is no closed-form solution of the likelihood, and when data augmentation algorithm speeds significantly the computation (for example, by avoiding expensive differential equation solvers).

Looking ahead, the OBDD model opens new avenues. **Uncoded** occurrences can often be attributed to specific subclades, and incorporating this information – similar to the use of topological constraints in traditional FBD analyses (Barido-Sottani et al., 2023) – would provide more direct insights into lineage-specific rates. In some cases, **uncoded** occurrences might be identified at the morphospecies level, offering even finer resolution (Silvestro et al., 2014; Stadler et al., 2018). **Additionally, model accuracy may be enhanced by placing an informative prior on the coding probability p (the proportion of fossil finds that are coded into the phylogeny), for instance assuming temporal autocorrelation, rather than letting it vary freely across time periods.** The fossil sampling

rates could also be allowed to evolve gradually along branches of the phylogeny, capturing variations in sampling across species and extending existing models with heterogeneous sampling processes (Barido-Sottani and Morlon, 2025). Statistically incorporating taphonomic empirical knowledge (e.g., the number of sites where the focal clade could have been found but was not) into the sampling rate priors could also be an improvement, as they provide direct, independent information about expected sampling intensity over time. These developments augur an exciting trajectory for models that, with ever-greater data integration and flexibility, can gradually uncover more subtle complexities of past evolutionary processes.

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CONFLICT OF INTEREST

The authors report no conflict of interest.

DATA AVAILABILITY STATEMENT

The code and data for reproducing the analyses are available at
<https://doi.org/10.5281/zenodo.16992580> and
<https://github.com/Jeremy-Andreoletti/OBDD>.

AUTHORS' CONTRIBUTIONS

J.A. and H.M. designed the project. I.Q. contributed the initial BDD and FBDD codes. J.A. wrote the OBDD code, performed the analyses, and wrote the first draft of the paper. All authors discussed the research and contributed to the writing and editing of the article.

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APPENDIX

A. LIKELIHOOD OF THE COMPLETE TREE

Following Quintero et al. (2025) for the FBDD model, we approximate the likelihood of a complete tree Ψ under the OBDD model by generating augmented discrete GBM paths along branches of the complete trees. Using the same notations for consistency, we set a small time step $\delta = t_{i+1} - t_i$ and divide each branch of the **reconstructed** tree into m steps such that $m = \lfloor t_b / \delta \rfloor + 1$, where t_b represents the edge length of branch b . For a **reconstructed-tree** branch b , which starts at t_s and ends at t_f , we sample the data augmented diffusion process at times $t = \{t_1 = t_s, t_2 = t_1 + \delta, \dots, t_{m+1} = t_f\}$. We thus obtain the stochastic evolution of speciation $\Lambda_b = \{\lambda_b(t_1), \dots, \lambda_b(t_{m+1})\}$ and of extinction $M_b = \{\mu_b(t_1), \dots, \mu_b(t_{m+1})\}$ in a discrete time grid. Thereafter $\lambda_l(t_i)$ is simplified as λ_i (and thus $\lambda_l(t_b) = \lambda_b$) and $\mu_l(t_i)$ as μ_i .

The combined likelihood for the complete tree Ψ (including internal branches Ψ_I and extinct branches Ψ_{T_μ}) and record of **uncoded** occurrences Ω given the OBDD model parameters as well as the branch-specific speciation and extinction rates, Λ and M respectively, is the following:

$$l(\Psi, \Omega | \Lambda, M, \lambda_0, \mu_0, \psi, \omega, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma_\mu) = l(\Psi | \Lambda, M, \lambda_0, \mu_0, \psi, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma_\mu) \times l(\Omega | \Psi, \omega) =$$

$$\prod_{p \in P} f_p \psi_p \prod_{b \in \Psi_I} \lambda_b \prod_{b \in \Psi_{T_\mu}} \mu_b \prod_{b \in \Psi} \prod_{i=1}^m p(\text{no event on the tree in } [t_i, t_{i+1}], \lambda_{i+1}, \mu_{i+1} | \lambda_i, \mu_i, \psi_p, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma_\mu)$$

$$\times \prod_{i \in I} e^{-\omega_i L_i N_i} \frac{(\omega_i L_i N_i)^{k_i}}{k_i!},$$

with f_p the number of **coded** fossils in the period p and k_i the number of **uncoded** occurrences in each interval i delimited both by the P periods of sampling rate change and the times of change in the diversity-through-time curve, such that in each interval of length L_i the **uncoded** occurrence sampling rate ω_i and the number of lineages N_i are constant.

As in Quintero et al. (2025), the probability that no event happens on the tree

during time $\delta = t_{i+1} - t_i$ and that the rates starting at λ_i and μ_i end up at λ_{i+1} and μ_{i+1} can be approximated as:

$$p(\text{no event on the tree in } [t_i, t_{i+1}], \lambda_{i+1}, \mu_{i+1} | \lambda_i, \mu_i, \psi_p, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma_\mu) = \\ \exp \left\{ -(\bar{\lambda}_{i,i+1} + \bar{\mu}_{i,i+1} + \psi_p)\delta \right\} \times \frac{1}{\sigma_\lambda \sqrt{2\pi\delta}} \exp \left\{ -\frac{(\Delta\lambda_i - \alpha_\lambda\delta)^2}{2\delta\sigma_\lambda^2} \right\} \times \frac{1}{\sigma_\mu \sqrt{2\pi\delta}} \exp \left\{ -\frac{(\Delta\mu_i - \alpha_\mu\delta)^2}{2\delta\sigma_\mu^2} \right\}.$$

B. MCMC RUNTIME

The running time for MCMC inferences under the OBDD model was measured across the 500 simulated datasets from the validation analysis and plotted as a function of the number of tips (including fossil tips). The results are shown in Figure 4.

We then applied a Generalized Additive Model for Location, Scale, and Shape (GAMLSS), using the *gamlss* package (Rigby and Stasinopoulos, 2005; R Core Team, 2018), to model the running time as a function of the number of extant tips, the number of fossils, and the number of **uncoded** occurrences. The GAMLSS framework allows us to model both the mean and variance structures separately, thus dealing with heteroscedasticity. We used a Gamma distribution for the response variable (running time), which is appropriate for modelling positive continuous data. The mean structure was modelled using a penalized B-spline (P-spline) for the number of extant tips and linear terms for the number of fossils and **uncoded** occurrences. Additionally, the scale (variance) of the model was allowed to depend on the number of extant tips. Only trees with at least 15 tips were retained in order to obtain normally distributed residuals and correct diagnostic plots.

Here is the resulting summary:

```
> summary(gamlss_model)
```

```
*****
```

```
Family:  c("GA", "Gamma")
```

```

808 Call:  gamlss(formula = running_time ~ pb(ntipsextant) + nfossils +
809         nocc, sigma.formula = ~ntipsextant, family = GA(mu.link = "identity",
810         sigma.link = "log"), data = subset(parameters_OBDD, par == "lambda" & ntips > 15))
811
812 Fitting method: RS()
813
814 -----
815 Mu link function:  identity
816 Mu Coefficients:
817
818             Estimate Std. Error t value Pr(>|t|)
819 (Intercept)    0.2627590   0.0287025   9.155 8.15e-16 ***
819 pb(ntipsextant) 0.0056901   0.0002565  22.183 < 2e-16 ***
820 nfossils       -0.0009767   0.0005983  -1.632   0.105
821 nocc           0.0001322   0.0001147   1.152   0.251
822 ---
823 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
824
825 -----
826 Sigma link function:  log
827 Sigma Coefficients:
828
829             Estimate Std. Error t value Pr(>|t|)
829 (Intercept) -0.2930249   0.0756514  -3.873 0.000165 ***
830 ntipsextant -0.0075807   0.0007572 -10.011 < 2e-16 ***
831 *****

```

832 While the number of extant tips has a significant non-linear effect on the running
833 time, we found no significant relationship between the running time and the number of
834 fossils or number of [uncoded](#) occurrences. These results indicate that tree size mostly

determines the computational burden and that including fossils have a negligible impact.

C. CETACEAN **UNCODED** OCCURRENCES OVER TIME

Occurrence ages are sampled uniformly within their age uncertainty ranges at the beginning of each analysis. Figure C.1 represents their temporal distribution for the 5 random seeds that were used.

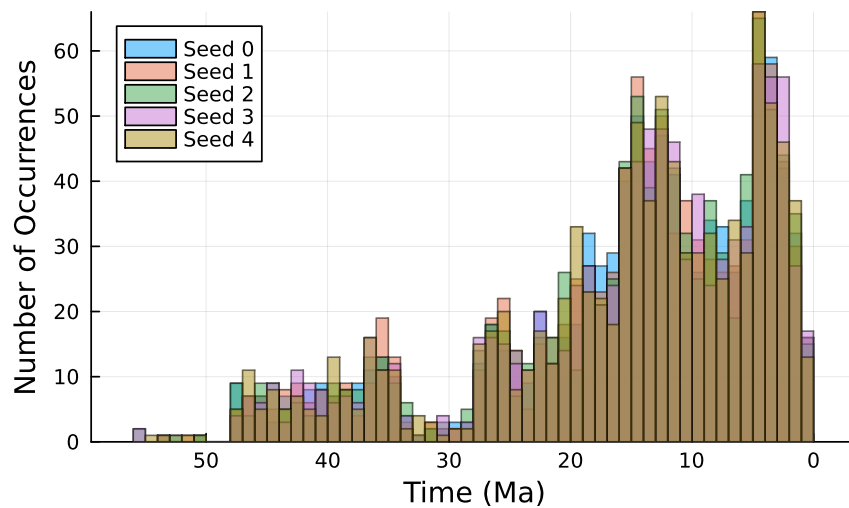


Figure C.1. **Temporal distribution of cetacean **uncoded** occurrences with age uncertainty.** These histograms illustrate the temporal distribution of cetacean **uncoded** occurrences over the Cenozoic. Results are overlaid for the five different sampling seeds used to uniformly sample within the age uncertainty ranges of the occurrences.

D. VALIDATION OF OBD IMPLEMENTATION

Our simulation-based calibration analysis of the Occurrence Birth-Death (OBD) model demonstrates a well-calibrated implementation for all parameters (Fig. D.1). The recovery proportions of the true parameter values consistently match the credible interval widths (D.1a) and the estimation errors decrease with increasing tree size (D.1b).

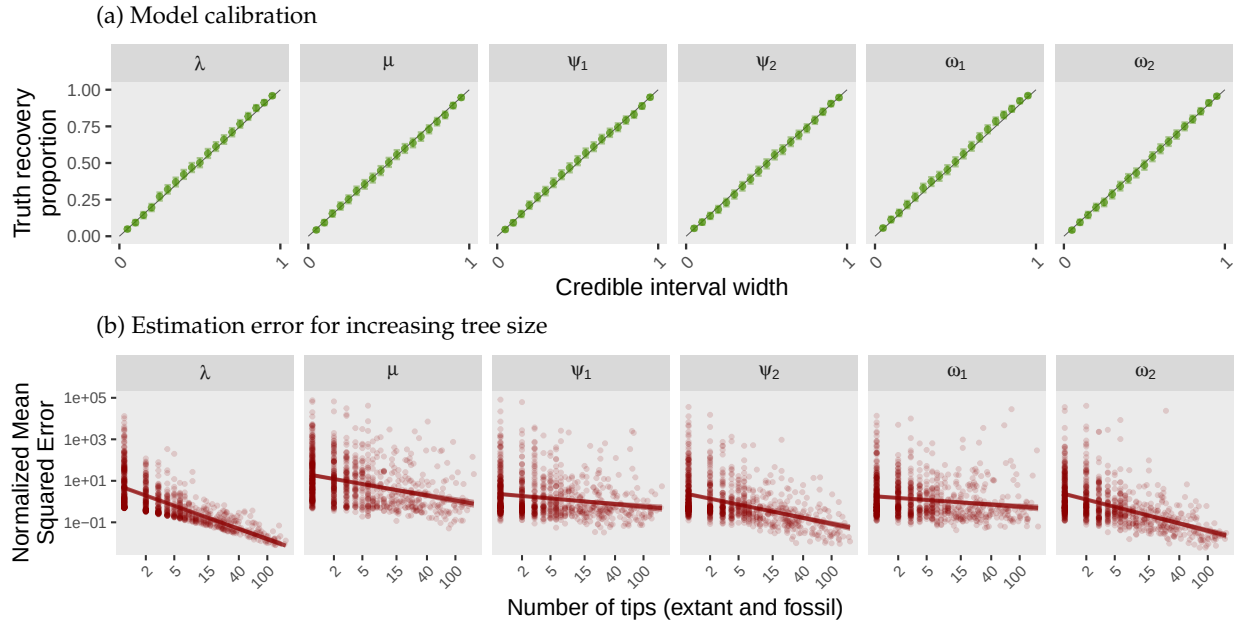


Figure D.1. **Simulation-based validation of the OBD model.** Results of the simulation-based calibration study on 1000 datasets simulated under the OBD model. (a) Truth recovery proportions: the proportion of datasets for which the true parameter value falls within posterior credible intervals of increasing width (from 5% to 95%). A well-calibrated model produces points along the diagonal. (b) Estimation error, measured as the normalized mean squared error NMSE $(\hat{\theta} - \theta)^2 / \theta^2$ (where $\hat{\theta}$ is the posterior median and θ the true value), plotted against the number of tips in the reconstructed phylogeny.

E. DRIVERS OF ESTIMATION ERROR

To explore the drivers of estimation error in the OBDD model parameters, we employed a generalized linear model (GLM) with the *stats* package in R (R Core Team, 2018) to predict the Normalized Mean Squared Error (NMSE) for each parameter based on some characteristics of the simulated datasets from the validation analysis. These predictors are the number of extant tips, the number of **coded** fossils, the number of **uncoded** occurrences, and the interaction between fossils and occurrences. We extracted estimated coefficients from these models and calculated the predicted mean influence of each predictor on the estimation error.

The analysis covered the following OBDD model parameters:

- λ_0 : Initial speciation rate.
- μ_0 : Initial extinction rate.
- α_λ : Speciation rate trend.
- α_μ : Extinction rate trend.
- σ_λ : Diffusion variance for speciation.
- σ_μ : Diffusion variance for extinction.
- ψ_1, ψ_2 : Fossil sampling rates in 2 epochs.
- ω_1, ω_2 : **Uncoded** occurrence sampling rates in 2 epochs.

The log(NMSE) for each parameter was modelled as a function of the logarithm of the number of extant tips ($\log(\text{ntipsextant})$), the square root of the number of fossils ($\sqrt{\text{nfossils}}$), the square root of the number of **uncoded** occurrences ($\sqrt{\text{noocc}}$), and the interaction between the number of **coded** fossils and **uncoded** occurrences ($\sqrt{\text{nfossils}} : \sqrt{\text{noocc}}$). There may be 0 **uncoded** occurrences or **coded** fossils, so a logarithmic transformation was not appropriate. For each parameter, we multiplied the coefficients by the average value of the predictors to calculate their predicted influence on estimation error, and display the resulting prediction in Figure E.1.

Figure E.2 complements the NMSE analysis above by showing the absolute estimation error (posterior median minus true value) for each OBDD parameter as a function of tree size. The LOESS curves are centred near zero for most parameters, confirming the absence of systematic bias. The scatter of individual estimates narrows markedly with increasing tree size, consistent with the NMSE results. The initial diversification rates (λ_0, μ_0) exhibit a slight negative bias, particularly for small trees, indicating mild underestimation of these rates when limited data cannot move the prior distributions.

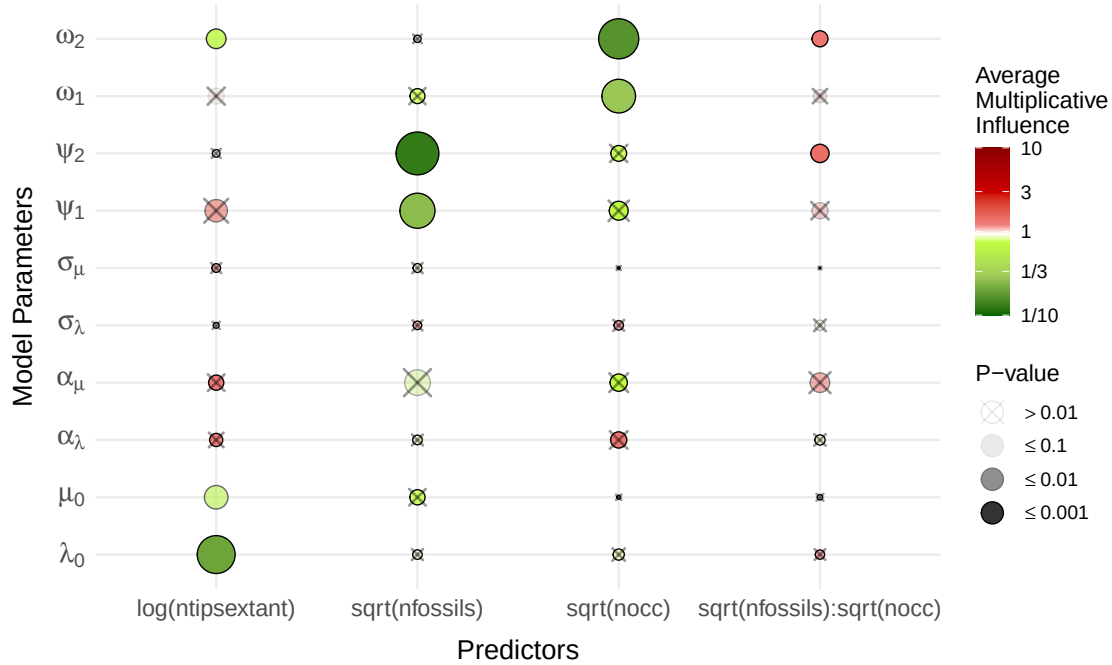


Figure E.1. **Drivers of estimation error for OBDD parameters.** This figure displays the drivers of estimation error, measured as the Normalized Mean Squared Error (NMSE). Each column corresponds to a predictor in a Generalized Linear Model: log of the extant tree size, square root of the number of fossils in the tree, fossils occurrences, and their interaction. Row correspond OBDD model parameters: initial speciation rate (λ_0), initial extinction rate (μ_0), speciation and extinction trends (α_λ , α_μ), diffusion variances (σ_λ , σ_μ), and fossil and occurrence sampling rates for two time intervals (ψ_1 , ψ_2 , ω_1 , ω_2). The colour of each circle indicates the average multiplicative influence of a predictor on a specific parameter's estimation error, and the circle size also reflects the absolute influence in log scale ($\times 10$ has a similar influence as $\times 1/10$). The shading of each circle reflects the statistical significance (p-value), with darker circles indicating lower p-values (more significant). Predictors with p-values above 0.01 are crossed.

To investigate whether the magnitude of rate variation influences parameter estimation, we plotted the NMSE of all global parameters against the true net diversification trend $\alpha_\lambda - \alpha_\mu$ (Fig. E.3). The LOESS curves are essentially flat in all panels, indicating that neither the direction nor the magnitude of the trend parameters has a detectable influence on estimation accuracy.

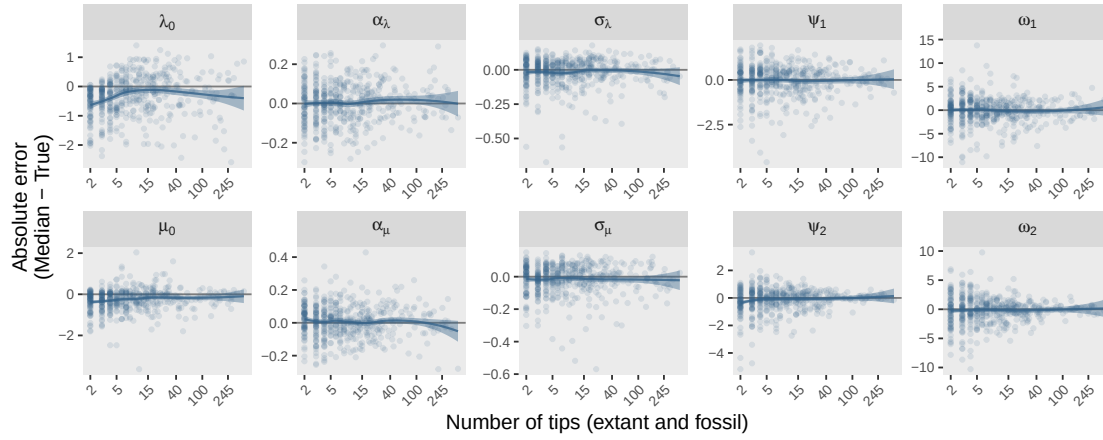


Figure E.2. **Absolute estimation error as a function of tree size for OBDD parameters.** Each panel displays the absolute error (posterior median – true value) for a model parameter as a function of the total number of tips (extant and fossil) on a log scale, across the 500 simulated datasets. Grey points represent individual datasets, and a LOESS curve (blue) with 95% confidence band summarizes the trend. Parameters shown are: initial speciation rate (λ_0), initial extinction rate (μ_0), speciation and extinction trends (α_λ , α_μ), diffusion variances (σ_λ , σ_μ), and fossil and occurrence sampling rates for two time intervals (ψ_1 , ψ_2 , ω_1 , ω_2). LOESS curves near zero indicate unbiased estimation; note the negative bias for diffusion variances and the large scatter in sampling rates for small trees.

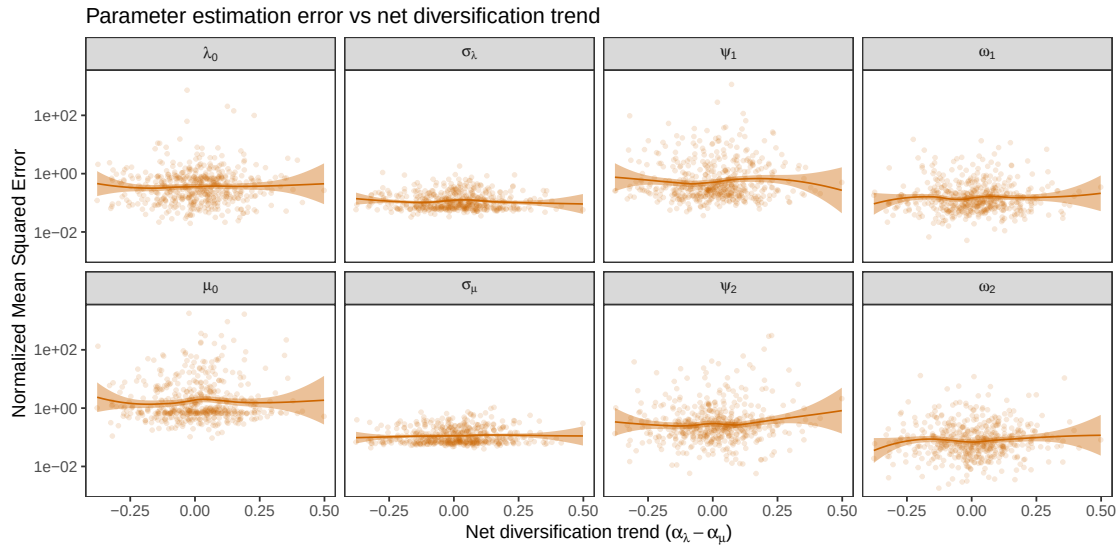


Figure E.3. **Estimation error as a function of the true net diversification trend $\alpha_\lambda - \alpha_\mu$.** Each panel displays the Normalized Mean Squared Error (NMSE) for a model parameter as a function of the true net diversification trend $\alpha_\lambda - \alpha_\mu$ across the 500 simulated OBDD datasets. Orange points represent individual datasets, and a LOESS curve summarizes the trend. The flat LOESS curves indicate no relationship between the speciation trend and estimation accuracy.